Specifications and Interactions in OLS

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Session 5: OLS Specification

Goals

1. What are our different model specification choices in OLS?
2. How do we choose between the different models?
3. How do we make inferences for interactions in OLS?

Model Specification

Even if you have only a handful of predictor variables to choose from, there are infinitely many ways to specify the right hand side of a regression. How do you decide what variables to include?

- Theory! (Most important, and nothing else is even close)
- T-test
- Model fit
- Bias? (look for changes in coefficient size/direction because of a variable's inclusion.

Theory

When should we include a control variable?

- If hypothesis testing is our goal, we only include control variables that confound X and Y.
- If prediction is our goal, we may want to include any variable associated with Y.
- Note that "Kitchen sink" regressions can reduce regression precision and even give misleading results.

Functional Form

While regression tests the linear relationships between variables, we can examine non-linear relationships as well.

- Linear-log
- Log-linear
- Log-log
- Curvilinear/Polynomial
- Interactions
- Lagged variables

Linear-log

Take the natural log of independent variable? Why?

- Positively (right) skewed
- Decreasing marginal effects
  - $y = \alpha + \beta ln(x) + \epsilon$
- Note: x must be a positive variable
- Interpret?: Divide the coefficient by 100. This tells us that a 1% increase in the independent variable increases (or decreases) the dependent variable by (coefficient/100) units. Example: the coefficient is 0.198. 0.198/100 = 0.00198. For every 1% increase in the independent variable, our dependent variable increases by about 0.002.

Log-linear

Take the natural log of dependent variable? Why?

- Positively (right) skewed
- If a variable is growing at an unknown rate, you may estimate this rate by regressing the natural log of the growing variable against time.
  - $ln(y) = \alpha + \beta x + \epsilon$
- Note: y must be a positive variable
- Interpret?: Exponentiate the coefficient, subtract one from this number, and multiply by 100. This gives the percent increase (or decrease) in the response for every one-unit increase in the independent variable. Example: the coefficient is 0.198. $(exp(0.198) – 1) * 100 = 21.9$. For every one-unit increase in the independent variable, our dependent variable increases by about 22%.

Log-log

Take the natural log of dependent variable and independent variable.

- This form is very popular for estimating production and demand functions
  - $ln(y) = \alpha + \beta ln(x) + \epsilon$
- Betas are unbiased estimators of the elasticity of Y with respect to the independent variables
- Interpret the coefficient as the percent increase in the dependent variable for every 1% increase in the independent variable. Example: the coefficient is 0.198. For every 1% increase in the independent variable, our dependent variable increases by about 0.20%.

Polynomial

We can add polynomial terms that allow the relationship between X and Y to be nonlinear

- This form is very popular for curvilinear effect. Examples, effect of democracy on terrorism or civil war
  - $y = \alpha + \beta_1 x^2 + \beta_2 x + \epsilon$

Adding a squared term in R

```
> ajr.mod <- lm(loggdp ~logmort0 + I(logmort0^2), data = ajr2)
>
> summary(ajr.mod)

Call:
lm(formula = loggdp ~ logmort0 + I(logmort0^2), data = ajr2)

Residuals:
     Min       1Q   Median       3Q      Max
-2.80046 -0.44980  0.04203  0.55290  1.48262

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   12.32661    1.00110  12.313  < 2e-16 ***
logmort0      -1.29152    0.41994  -3.076  0.00314 **
I(logmort0^2)  0.07459    0.04269   1.747  0.08562 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7604 on 61 degrees of freedom
Multiple R-squared:   0.49,Adjusted R-squared:  0.4732
F-statistic:  29.3 on 2 and 61 DF,  p-value: 1.207e-09
```

Adding a squared term

Lagged independent variables

Imagine that X affects Y with some instantaneous effect, plus an effect that appears after time has passed? Different lag structure can lead to different dynamic inferences

- $y = \beta_0 y_{t-1} + \beta_1 x_t + \beta_2 x_{t-1} + \epsilon$
- $\Delta y = \beta_0 y_{t-1} + \beta_1 \Delta x_t + \beta_2 x_{t-1} + \epsilon$

Interaction

What if one independent variable's effect upon y depends upon the level of a different independent variable?

- This form is very popular for applied research: When does democracy affect conflict? When does gender affect vote choice?
  - $y = \alpha + \beta_1 x + \beta_2 w + \beta_3 x * w + \epsilon$

Note well: Always include the marginal effects (sometimes called the lower order terms or additive terms)

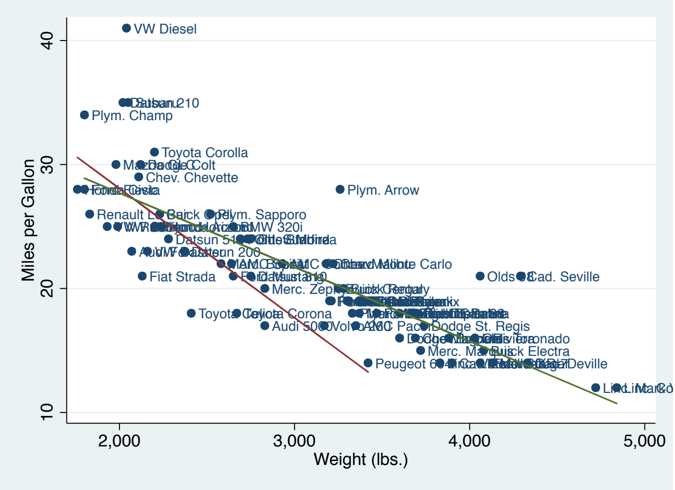
MPG Regression with Foreign Dummy & Interaction

	(1)	(2)
	mpg	mpg
Weight	-0.007***	-0.006***
	(0.001)	(0.001)
Foreign	-1.650	9.271**
	(1.076)	(4.500)
Weight $*$ Foreign		-0.004**
		(0.002)
Constant	41.680***	39.647***
	(2.166)	(2.243)
Observations	74	74
R-squared	0.663	0.690

Standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.1$

Dummy Variables: MPG and Weight



Interaction: Two lines in one regression

$y = \beta_0 + \beta_1 x + \beta_2 w + \beta_3 x * w + \epsilon$

- Consider w as binary. When w=0:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- When w=1:

$$\hat{y} = (\hat{\beta}_0 + \hat{\beta}_2) + (\hat{\beta}_1 + \hat{\beta}_3) x$$

Interaction: Interpretation

$y = \beta_0 + \beta_1 x + \beta_2 w + \beta_3 x * w + \epsilon$

- β_0 : average value of y when both x and w are equal to 0
- β_1 : a one-unit change in x is associated with a β_1 - unit change in y when w = 0 (note the difference here with MV regression).
- β_2 : average difference in y between w = 1 group and w = 0 group when x = 0
- β_3 : change in the effect of x on y between w = 1 group and w = 0
- Null hypothesis for β_3: no change in effect of x (null is NOT that there is not effect for β_3).

Interaction: when w is continuous

Same model:

$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x * w + \epsilon$

- With a continuous w , we can have more than two values that it can take on:

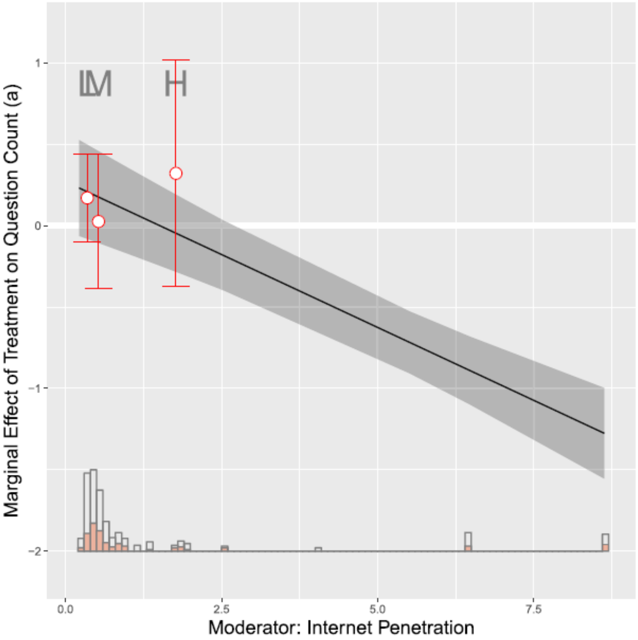
	Intercept for x_i	Slope for x_i
w=0	$\hat{\beta}_0 + \hat{\beta}_2 \times 0$	$\hat{\beta}_1 + \hat{\beta}_3 \times 0$
w=1	$\hat{\beta}_0 + \hat{\beta}_2 \times 1$	$\hat{\beta}_1 + \hat{\beta}_3 \times 1$
w=5	$\hat{\beta}_0 + \hat{\beta}_2 \times 5$	$\hat{\beta}_1 + \hat{\beta}_3 \times 5$

Interaction: Interpretation of continuous w

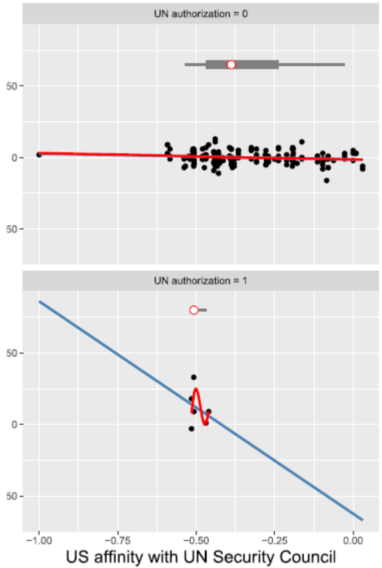
- β_0 : average value of y when both x and w are equal to 0
- β_1 : a one-unit change in x is associated with a β_1 - unit change in y when w = 0 (same as before).
- β_2 : a one-unit change in w is associated with a β_2 - unit change in y.
- β_3 : the change in the effect of x given a one-unit change in w, or
- β_3 : the change in the effect of w given a one-unit change in x (symmetric hypotheses).

Interaction problems!

- Standard errors are probably too generous (Type I errors)
- Constant effects (conditional on moderator)
- Lack of Common Support



Marginal Effect of Treatment on Question Count (a)
Moderator: Internet Penetration
L
M
H
1
0
-1
-2
0.0
2.5
5.0
7.5


UN authorization = 0
UN authorization = 1
50
0
50
−1.00
−0.75
−0.50
−0.25
0.00
US affinity with UN Security Council

How Much Should We Trust Estimates from Multiplicative Interaction Models? Simple Tools to Improve Empirical Practice

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[121] Dear Political Scientists: Don't Bin, GAM Instead

Posted on December 3, 2024 by Uri Simonsohn

Model Specification

- When do we include a variable in our model?
- Parsimony — include theoretically relevant variables.
- Do not fit noise!
  - Every additional explanatory variable you include will increase the percentage of the dependent variable explained.
  - Variables may appear statistically significant due to chance alone.
- Pay attention to your degrees of freedom!

Model Specification

- Check for potential multicollinearity.
- Are you omitting any variables that might cause y and be correlated with the other independent variables?
- How do our estimates change as we include and exclude variables?

Regression Decisions

The following are general guidelines for building a regression model

- Make sure all relevant predictors are included. These are based on your research question, theory and knowledge on the topic.
- Combine those predictors that tend to measure the same thing (i.e. as an index).
- Consider the possibility of adding interactions (mainly for those variables with large effects)
- Strategy to keep or drop variables:
    1. Predictor not significant and has the expected sign $\rightarrow$ Keep it
    2. Predictor not significant and does not have the expected sign $\rightarrow$ Drop it
    3. Predictor is significant and has the expected sign $\rightarrow$ Keep it
    4. Predictor is significant but does not have the expected sign $\rightarrow$ Review, you may need more variables, it may be interacting with another variable in the model or there may be an error in the data